

Dancing with Giants: Learning Goal-Directed Non-Verbal Attunement with Large-Scale Inflatable Robots

Ellen Weir

Faculty of Engineering
University of Bristol
Bristol, United Kingdom
ellen.weir@bristol.ac.uk

Abstract

This research investigates non-verbal, embodied interaction with unfamiliar agents, focusing on large-scale soft robotic systems and their potential to support learning, coordination, and self-reflection. Inspired by equine-assisted interventions (EAIs), therapeutic practices with live horses that foster emotional regulation and wellbeing, it examines interaction as a relational process shaped by ambiguity, resistance, and reciprocal responsiveness. Across autoethnography, co-design, and interaction studies, it explores how attunement emerges through breakdown, repair, and embodied negotiation. Soft, inflatable robots provide a material platform that combines physical safety with human-scale presence, enabling exploration of autonomy, uncertainty, and embodied risk. Rather than replacing EAIs, this work investigates how relational mechanisms can be translated into robotic systems to support reflective awareness of bodily behaviour, coordination, and co-regulation. It contributes to HCI and HRI by challenging comfort-oriented design paradigms and positioning robots as relational partners shaping interaction through calibrated unpredictability and reciprocal embodiment.

CCS Concepts

• **Human-centered computing** → **Empirical studies in HCI**;
Empirical studies in collaborative and social computing.

Keywords

Human-Robot Interaction; Embodied interaction; Soft Robotics

ACM Reference Format:

Ellen Weir. 2026. Dancing with Giants: Learning Goal-Directed Non-Verbal Attunement with Large-Scale Inflatable Robots. In *Extended Abstracts of the 2026 CHI Conference on Human Factors in Computing Systems (CHI EA '26)*, April 13–17, 2026, Barcelona, Spain. ACM, New York, NY, USA, 4 pages. <https://doi.org/10.1145/3772363.3799216>

1 Introduction

This research investigates how people learn to communicate, coordinate, and reflect on their own behaviour through non-verbal, embodied interaction with unfamiliar agents, and how such processes may be supported by large, responsive soft robotic systems. The work is inspired by equine-assisted interventions (EAIs), therapeutic practices in which clients develop interpersonal skills, such

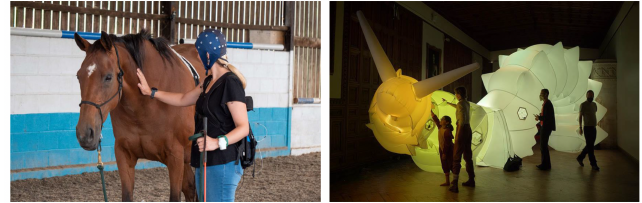


Figure 1: Left: A client engaging in an equine-assisted intervention, attending to the horse's posture, movement, and affective state in order to achieve coordinated motion. Right: Members of the public interacting with a human-scale inflatable soft robot, adjusting distance and bodily movement to explore non-verbal responsiveness. The juxtaposition illustrates how both contexts rely on whole-body communication rather than verbal instruction.

as emotional regulation and self-awareness, through sustained bodily interaction with horses rather than verbal instruction alone [5, 29].

Rather than framing interaction as command execution or task completion, this research approaches interaction as relational and emergent. Communication develops through ambiguity, resistance, adjustment, and reciprocal responsiveness. Through a programme of qualitative, embodied, and design-led studies, this work examines how attunement develops over time when humans interact with agents whose behaviour is not fully controllable or immediately legible. It then explores whether large, responsive inflatable robots can be designed to support comparable learning dynamics.

Importantly, this work does not propose a technological substitute for horses. Instead, EAIs are used as an analytical lens through which relational mechanisms of non-verbal coordination and co-regulation can be examined and abstracted. The focus is therefore on interactional processes and embodied learning, rather than on clinical therapeutic outcomes.

2 Research Context

Horses, as prey animals [8], exhibit a heightened sensitivity to non-verbal cues and the affective states of others [16]. This sensitivity is central to equine-assisted interventions, in which clients must learn to communicate non-verbally and regulate their emotional states in order to achieve effective interaction and eventual attunement with the horse [11, 16, 21, 28].

A defining feature of EAIs is the horse's perceived autonomy and unpredictability. Its scale, embodied presence, and capacity



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ACM ISBN 979-8-4007-2281-3/26/04

<https://doi.org/10.1145/3772363.3799216>

for refusal introduce a sense of risk that intensifies client awareness, empathy, and the need for emotional regulation before communication can occur [1, 6, 20]. Leadership within EAIs is fluid: excessive control prompts resistance, while passivity invites the horse to assume leadership. Emotional meaning emerges relationally, forming a dynamic dyad in which agency is distributed across human and non-human bodies. This entangled human–horse interaction positions horses as co-regulators within the relational system rather than passive instruments, highlighting how agency emerges through interaction [2].

Central to this process is attunement. In EAIs, attunement refers to a gradual increase in sensitivity and responsiveness between interaction partners [20], as participants learn to notice the horse’s signals, regulate their own internal state, and modify their behaviour in response (Figure 2 & 3). This includes coordination of movement rhythms, affective states, and shared attention, and is often described as a sense of mutual responsiveness or “feeling felt” [18, 23, 32]. Research across both human-human and human-animal interaction suggests that such synchrony is linked to stronger feelings of empathy, and mutual understanding [3, 32]. In EAIs, the development of attunement is closely connected to emotional self-regulation, as participants learn to adjust their affective energy and body language in order to maintain a calm and respectful interaction [9, 34].

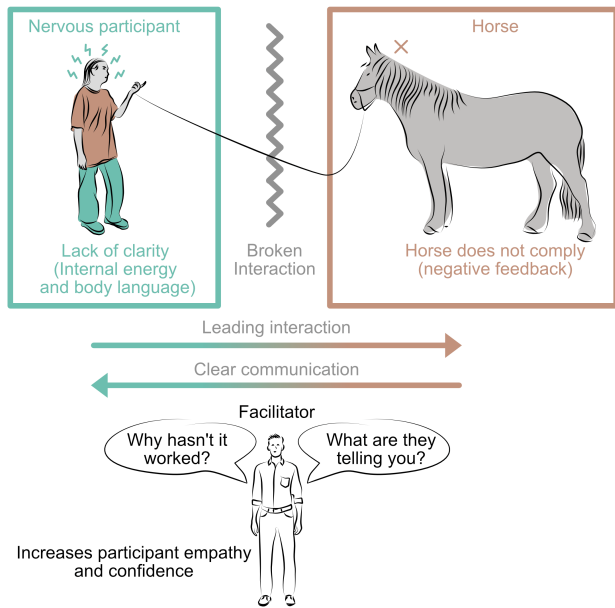


Figure 2: Early-phase disjointed interaction between EAI client and horse [33]. The client attempts to lead the interaction through unclear or dysregulated bodily cues, while the horse resists or fails to comply. Facilitator guidance supports the client in recognising how their own non-verbal behaviour shapes the interaction, prompting adjustment and increasing attunement.

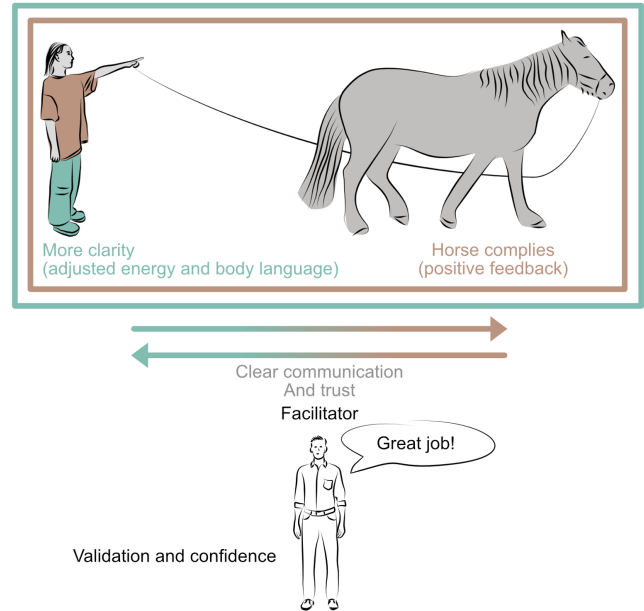


Figure 3: Developing attunement between client and horse [33]. As the client demonstrates clearer, more regulated body language and leadership, the horse responds with coordinated movement. The reciprocal exchange reinforces trust and shared rhythm, with the facilitator validating progress.

While EAIs have demonstrated benefits for individuals who struggle with talk-based therapies [10, 28], they remain resource-intensive and difficult to scale [25, 30]. This motivates the exploration of technological counterparts that can embody similar interactional mechanisms without attempting to replace the horse itself. Rather than prioritising task efficiency or verbal communication, this research examines how ambiguity, resistance, and embodied negotiation can support reflection and learning over time.

3 HRI Implications

Within HCI and HRI, robots are increasingly designed to support emotional and social engagement beyond purely functional roles [7, 19]. However, much existing work continues to emphasise predictability, legibility, and comfort, often framing robots as compliant companions or tools [24].

Recent HRI research has begun to question comfort-oriented paradigms, suggesting that interpretable unpredictability and resistance can play a productive role in supporting reflection and self-awareness [14, 33]. Jain and Gardner McCune for example argued that “the horse, in many ways, is the kind of teammate and companion that is the holy grail for co-bots and social robots” [15]. Extending this perspective, Weir et al. proposed that therapeutic robotics could directly draw on the mechanisms underpinning EAIs, positioning robots as relational partners with meaningful agency [33], to prompt users to adapt their behaviour, regulate emotion, and develop empathy [1, 6].

Soft robotics offers a particularly suitable material platform for investigating these ideas. Inflatable and deformable systems are

typically experienced as physically safe and approachable [22, 27]; however, when realised at human scale they can also influence bodily movement, limit action, and provoke notable affective reactions [4]. Despite these qualities, large-scale soft robots remain comparatively underexplored within HCI, especially regarding their potential to support learning, interpretation, and the emergence of relational agency over time.

4 Research Questions

This PhD investigates whether interactional mechanisms observed in equine-assisted interventions (EAIs) can inform the design of relational human–robot interaction, and whether such interactions can support reflection, coordination, and emotional self-regulation.

The research is guided by the following overarching question: **How can mechanisms of attunement and co-regulation observed in equine-assisted interventions be abstracted and implemented within non-verbal human–robot interaction?**

- **RQ1:** What mechanisms within EAIs contribute to the associated therapeutic outcomes, and what factors support these?
- **RQ2:** What behavioural, material, and interactional properties must a robot possess in order to support these mechanisms; and what constraints and opportunities arise in attempting such translation?
- **RQ3:** What are the outcomes for the user when interacting with such a robot informed by EAI mechanisms?
- **RQ4:** What methodological approaches are most appropriate for investigating relational, agentic, and potentially risky interactions across novel robotic design contexts?

Together, these questions frame the thesis as a progression from observation, to translation, to evaluation. The aim is not to replicate EAIs, but to investigate whether selected relational properties of embodied interaction with a responsive non-human agent can be supported by robotic systems.

5 Research Structure

This PhD advances a research programme investigating how people learn to communicate, coordinate, and reflect through repeated embodied interaction with unfamiliar non-verbal agents, and how these processes may be supported by large-scale soft robotic systems. Rather than treating ambiguity as a design flaw to be minimised, this work positions ambiguity as an interactional resource that invites curiosity and reflection.

The first study consists of an autoethnographic investigation of an equine-assisted intervention [33]. The first-person approach uses the researcher’s lived bodily experience as a resource for understanding interactional mechanisms [13, 31]. This study identified reciprocal responsiveness, resistance to dysregulated behaviour, embodied risk, and facilitator-supported sense-making as central features of the intervention.

The second study employed puppeteering and co-design as embodied design methods to translate EAI interaction principles into speculative robotic forms. These approaches operate as low-cost, low-fidelity exploratory design techniques [17, 26], enabling designers to enact and examine interaction before technical implementation. Through collaborative enactment with practitioners

from HCI, social work, puppetry, and equine-assisted interventions, this study explored how perceived agency and resistance could be intentionally designed.

The third study investigated first-time interactions with human-scale soft robotic systems, focusing on how form, proximity, and responsiveness influence perceptions of agency, safety, and engagement. Participants interacted with large inflatable robots exhibiting simple autonomous behaviour. Methodologically, the research is informed by embodied and somaesthetic perspectives in HCI [12, 13], which regard lived bodily experience as a legitimate form of knowledge production. Instead of assessing interaction primarily through task completion or conventional usability measures, the systems themselves acted as material probes, allowing processes of breakdown, hesitation, repair, and coordination to be examined empirically.

The final study extends this work through repeated interaction with LUMA (the robot shown in *Figure 1*), a large inflatable soft robot constructed entirely from flexible textile materials and actuated through air. This study examines how participants learn to interpret and respond to non-verbal robotic behaviour over time, and how coordination, confidence, and attunement develop through repeated engagement.

6 Ethical Considerations

This research engages with interactional uncertainty, embodied risk, and discomfort that accompanies this; which raises important ethical considerations. While physical safety is prioritised through the use of soft materials, the work intentionally explores calibrated affective vulnerability and temporary loss of control within ethically managed research conditions. The research also remains attentive to the ethical distinction between living animals and designed artefacts. While EAIs inform the design rationale, the work does not attempt to replicate or simulate animal sentience. Instead, it examines how specific interactional mechanisms can be meaningfully translated into robotic systems while avoiding deceptive anthropomorphism.

7 Future Work

Together, these studies establish a conceptual and empirical foundation for a final investigation into learning and attunement, through repeated interaction with a large-scale soft robotic agent. At present, the work remains exploratory, focusing on how people initially perceive, interpret, and physically negotiate interaction with an unfamiliar non-verbal system, and on what these encounters reveal about relational design in HRI.

The aim is to position the robot neither as a tool to be mastered nor as a companion designed for comfort, but as a relational artefact whose value emerges through effortful interaction. By empirically examining reciprocity and attunement over time, the research will address questions regarding the limits of predictability, the role of resistance in fostering reflection, and the conditions under which ambiguity remains productive rather than overwhelming.

This work lays the groundwork for future investigations into whether selected relational and regulatory mechanisms observed in equine-assisted interventions might be meaningfully supported by large-scale soft robotic systems, while remaining attentive to

the ethical, experiential, and ontological differences between living beings and designed artefacts.

Future research will extend beyond first encounters toward situated and sustained use. In particular, it will be important to identify which user groups may most meaningfully benefit from such interactions, and appropriate application contexts, including different forms of therapeutic, wellbeing, or reflective practice in which non-verbal interaction may complement rather than replace existing approaches. Further work will also investigate how embodiment shapes outcomes. While the present research focuses on a large inflatable soft robot, alternative embodiments may afford different interactional dynamics.

Ultimately, the aim is not to prescribe a single application but to understand the design implications of relational interaction itself.

Acknowledgments

This research programme was made possible through the guidance and support of Professor Anne Roudaut and Professor Ute Leonards; Harriet Laurie MBE, the Founder of the studied Equine-assisted intervention; and Richard Sewell, the Director of Pneumatics at Airgiants. This work is supported by the Engineering and Physical Sciences Research Council (EP/S023704/1) and the UK Research and Innovation Strength in Places Fund. MyWorld and The Brigstow Institute are also gratefully acknowledged for their funding and support, and Matt Sutton for creating the images used in this work.

References

- [1] Keren Bachi. 2013. Application of attachment theory to equine-facilitated psychotherapy. *Journal of Contemporary Psychotherapy* 43 (2013), 187–196.
- [2] Karen Barad. 1996. Meeting the universe halfway: Realism and social constructivism without contradiction. In *Feminism, science, and the philosophy of science*. Springer, 161–194.
- [3] Andrea Behrends, Sybille Müller, and Isabel Dziobek. 2012. Moving in and out of synchrony: A concept for a new intervention fostering empathy through interactional movement and dance. *The arts in Psychotherapy* 39, 2 (2012), 107–116.
- [4] Bijetri Biswas, Emma Powell, Robert Nixdorf, Richard Sewell, and Anne Roudaut. 2025. Encounter with the Giants: Understanding Interaction with Large-scale Inflatable Soft Robots. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*. 1–15.
- [5] Marta Borgi, Dafne Loliva, Stefania Cerino, Flavia Chiarotti, Aldina Venerosi, Maria Bramini, Enrico Nonnis, Marco Marcelli, Claudia Vinti, Chiara De Santis, et al. 2016. Effectiveness of a standardized equine-assisted therapy program for children with autism spectrum disorder. *Journal of autism and developmental disorders* 46 (2016), 1–9.
- [6] Hannah Burgon, Di Gammage, and Jenny Hebden. 2018. Hoofbeats and heartbeats: equine-assisted therapy and learning with young people with psychosocial issues—theory and practice. *Journal of Social Work Practice* 32, 1 (2018), 3–16.
- [7] EunJeong Cheon, Eike Schneiders, Kristina Diekjobst, and Mikael B Skov. 2022. Robots as a Place for Socializing: Influences of Collaborative Robots on Social Dynamics In-and Outside the Production Cells. *Proceedings of the ACM on Human-Computer Interaction* 6, CSCW2 (2022), 1–26.
- [8] Janne Winther Christensen, Linda Jane Keeling, and Birte Lindstrøm Nielsen. 2005. Responses of horses to novel visual, olfactory and auditory stimuli. *Applied Animal Behaviour Science* 93, 1–2 (2005), 53–65.
- [9] Andrea Guidi, Antonio Lanata, Paolo Baragli, Gaetano Valenza, and Enzo Pasquale Scilingo. 2016. A wearable system for the evaluation of the human-horse interaction: A preliminary study. *Electronics* 5, 4 (2016), 63.
- [10] Ann Hemingway. 2019. A study exploring the implementation of an equine assisted intervention for young people with mental health and behavioural issues. *J 2*, 2 (2019), 236–246.
- [11] Ann Hemingway and Kezia Sullivan. 2022. Reducing the incidence of domestic violence: An observational study of an equine-assisted intervention. *Family process* 61, 2 (2022), 549–570.
- [12] Kristina Höök. 2010. Transferring qualities from horseback riding to design. In *Proceedings of the 6th Nordic Conference on Human-Computer Interaction: Extending Boundaries*. 226–235.
- [13] Kristina Hook. 2018. *Designing with the body: Somaesthetic interaction design*. MIT Press.
- [14] Eakta Jain and J. Cortney Bradford. 2024. Rebellion and Disobedience in HRI: Insights from Human-Horse Interaction. In *Proceedings of the ACM HRI 2024 Workshop on Rebellion and Disobedience in Human-Robot Interaction (RAD-HRI)*. Association for Computing Machinery.
- [15] Eakta Jain and Christina Gardner-McCune. 2023. Horse as Teacher: How human-horse interaction informs human-robot interaction. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*. 1–13.
- [16] Kathleen Kemp, Tania Signal, Helena Botros, Nik Taylor, and Kathy Prentice. 2014. Equine facilitated therapy with children and adolescents who have been sexually abused: A program evaluation study. *Journal of child and family studies* 23 (2014), 558–566.
- [17] Scott R Klemmer, Björn Hartmann, and Leila Takayama. 2006. How bodies matter: five themes for interaction design. In *Proceedings of the 6th conference on Designing Interactive systems*. 140–149.
- [18] Sander L Koole and Wolfgang Tschacher. 2016. Synchrony in psychotherapy: A review and an integrative framework for the therapeutic alliance. *Frontiers in psychology* 7 (2016), 862.
- [19] Frank Krueger, Kelsey C Mitchell, Gopikrishna Deshpande, and Jeffrey S Katz. 2021. Human–dog relationships as a working framework for exploring human–robot attachment: a multidisciplinary review. *Animal Cognition* 24 (2021), 371–385.
- [20] H. Laurie and J. Noble. 2015. Evidence Reviews Theory of Change and Contribution Analysis. TheHorseCourse.
- [21] Jennifer A Lentini and Michele Knox. 2009. A qualitative and quantitative review of equine facilitated psychotherapy (EFP) with children and adolescents. *The open complementary medicine journal* 1, 1 (2009).
- [22] Carmel Majidi. 2014. Soft robotics: a perspective—current trends and prospects for the future. *Soft robotics* 1, 1 (2014), 5–11.
- [23] Mario Mikulincer, Phillip R Shaver, and Ety Berant. 2013. An attachment perspective on therapeutic processes and outcomes. *Journal of personality* 81, 6 (2013), 606–616.
- [24] Don Norman. 2013. *The Design of Everyday Things* (revised and expanded ed.). Basic Books, New York.
- [25] Jeffrey R Nurenberg, Steven J Schleifer, Thomas M Shaffer, Mary Yellin, Prital J Desai, Ruchi Amin, Axel Bouchard, and Cristina Montalvo. 2015. Animal-assisted therapy with chronic psychiatric inpatients: Equine-assisted psychotherapy and aggressive behavior. *Psychiatric services* 66, 1 (2015), 80–86.
- [26] David Porfirio, Evan Fisher, Allison Sauppe, Aws Albarghouthi, and Bilge Mutlu. 2019. Bodystorming human-robot interactions. In *proceedings of the 32nd annual ACM symposium on user Interface software and technology*. 479–491.
- [27] Daniela Rus and Michael T Tolley. 2015. Design, fabrication and control of soft robots. *Nature* 521, 7553 (2015), 467–475.
- [28] Nikki Savvides. 2012. Communication as a solution to conflict: Fundamental similarities in divergent methods of horse training. *Society & Animals* 20, 1 (2012), 75–90.
- [29] Fiona Seaman-Thornton. 2020. *Exploring change processes in an equine-assisted intervention*. Ph. D. Dissertation. Canterbury Christ Church University.
- [30] Mark Shucksmith, Jayne Glass, Jane Atterton, and Polly Chapman. 2023. *Rural poverty today: experiences of social exclusion in rural Britain*. Policy Press.
- [31] Dag Svanæs. 2013. Interaction design for and with the lived body: Some implications of merleau-ponty's phenomenology. *ACM transactions on computer-human interaction (TOCHI)* 20, 1 (2013), 1–30.
- [32] Wolfgang Tschacher, Georg M Rees, and Fabian Ramseyer. 2014. Nonverbal synchrony and affect in dyadic interactions. *Frontiers in psychology* 5 (2014), 1323.
- [33] Ellen Weir, Ute Leonards, and Anne Roudaut. 2025. “You Can Fool Me, You Can’t Fool Her!”: Autoethnographic Insights from Equine-Assisted Interventions to Inform Therapeutic Robot Design. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*. 1–20.
- [34] Jan Yorke, William Nugent, Elizabeth Strand, Rebecca Bolen, John New, and Cindy Davis. 2013. Equine-assisted therapy and its impact on cortisol levels of children and horses: A pilot study and meta-analysis. *Early child development and care* 183, 7 (2013), 874–894.